

MOVING CHARGES AND MAGNETISM

Chapter Four
Class - XII

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Learning Objectives:

- Concept of magnetic field, Oersted's experiment.
- Biot - Savart law and its application to current carrying circular loop.
- Ampere's law and its applications to infinitely long straight wire. Straight solenoid (only qualitative treatment).
- Force on a moving charge in uniform magnetic and electric fields.
- Force on a current-carrying conductor in a uniform magnetic field, between two parallel current-carrying conductors-definition of ampere,

Oersted's Experiment

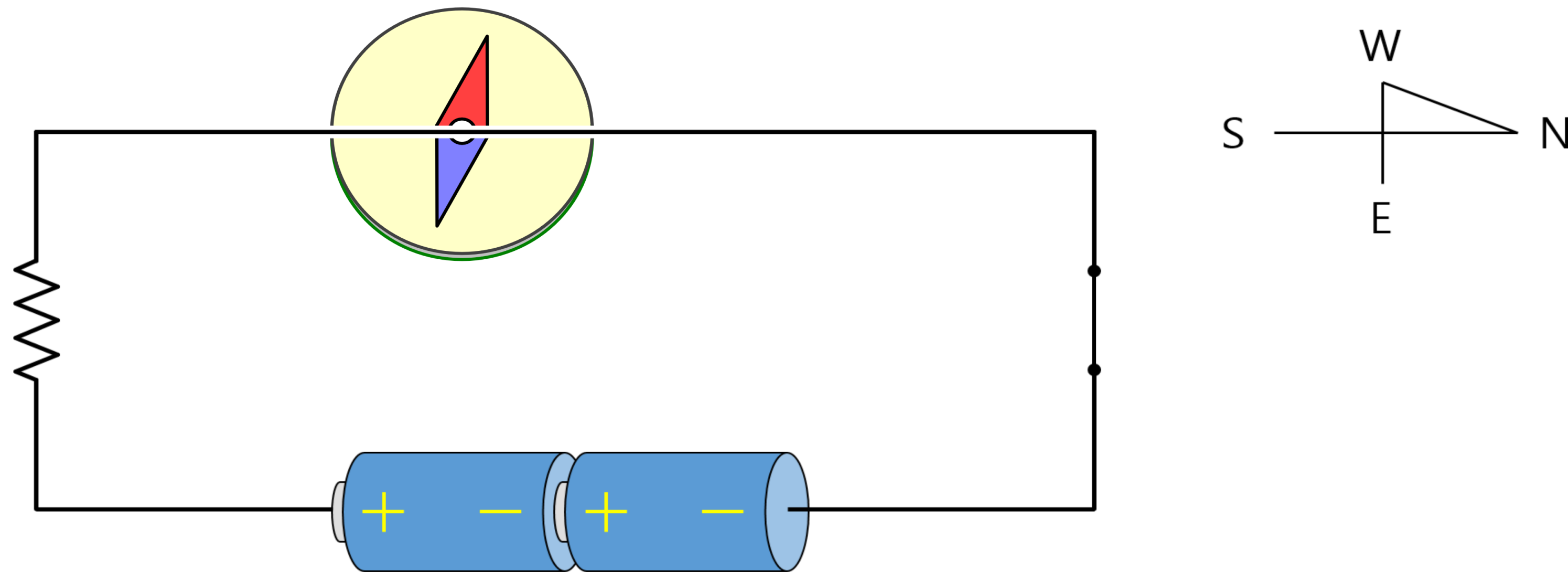


Figure: Oersted's Experiment Setup.
Source : Java Lab

- Danish physicist **Hans Christian Oersted** **accidentally discovered** that a **current in a straight wire caused** a noticeable **deflection** in a nearby **magnetic compass needle**.
- **Oersted experimented** with this phenomenon several times, **revealing a close relationship between electrical and magnetic phenomena**.

Oersted's Experiment

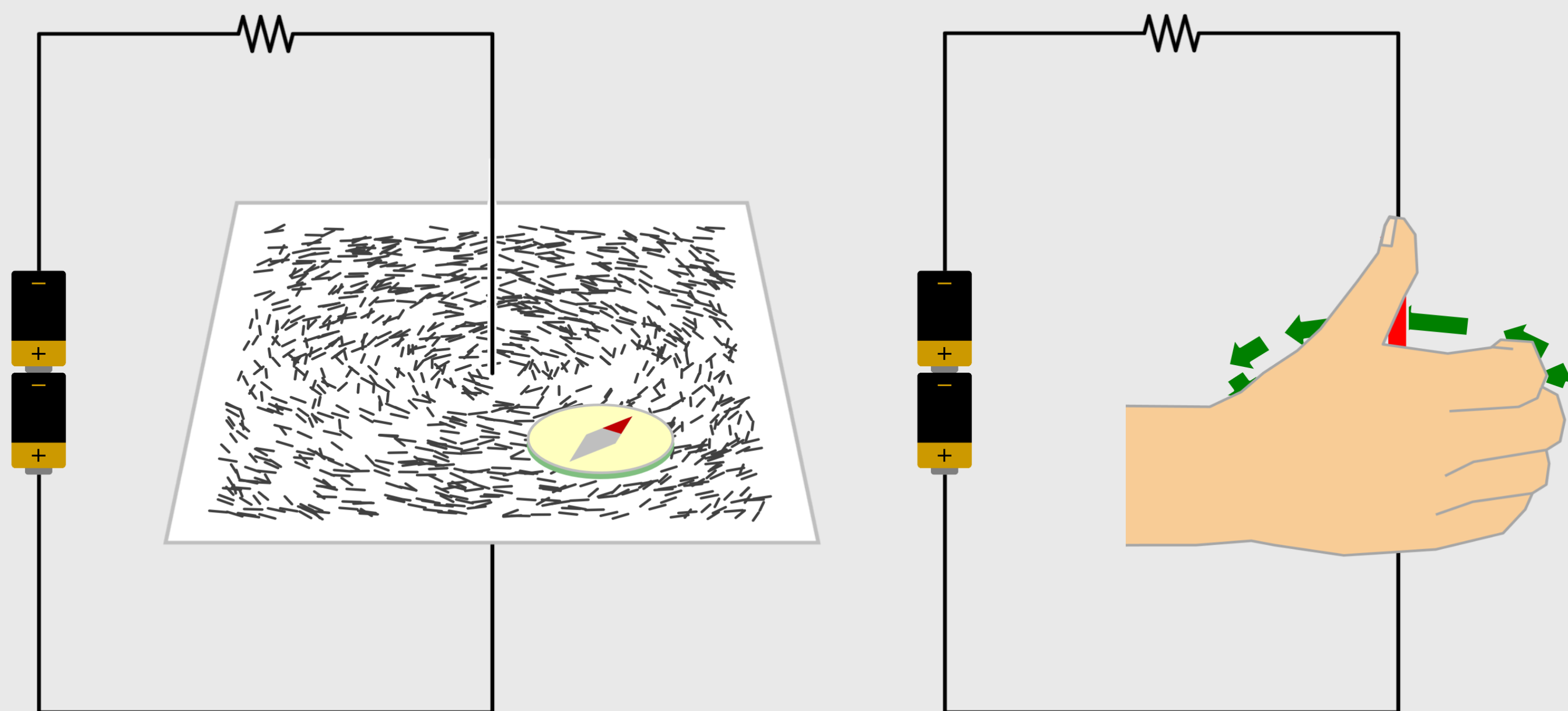


Figure: Magnetic Field around a Wire.
Source : Java Lab

Observations and Results:

- He found that the **alignment of the needle is tangential to an imaginary circle** which has the straight wire as its center and has its plane perpendicular to the wire.
- **Reversing** the direction of the **current reverses the orientation of the needle.**
- The **deflection increases on increasing the current or bringing the needle closer to the wire.**

Oersted's Experiment

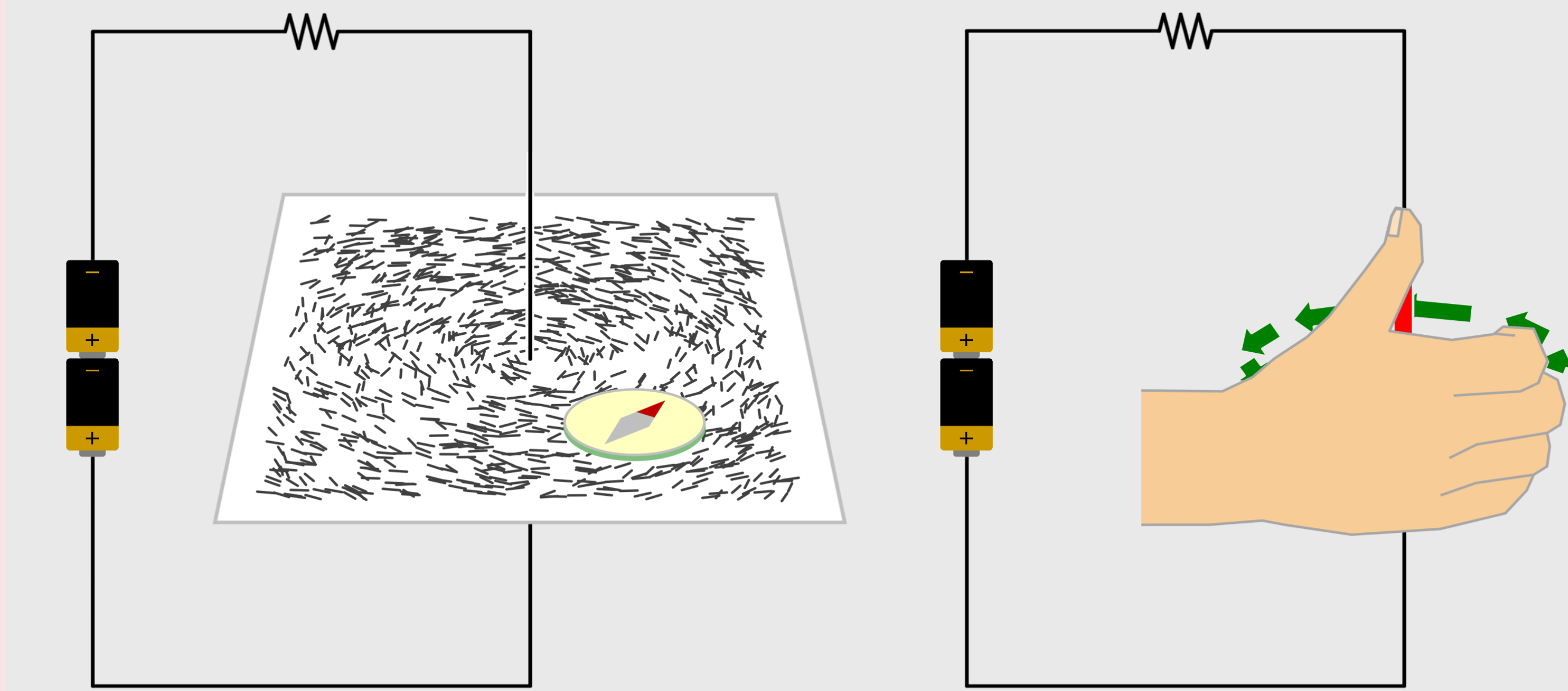
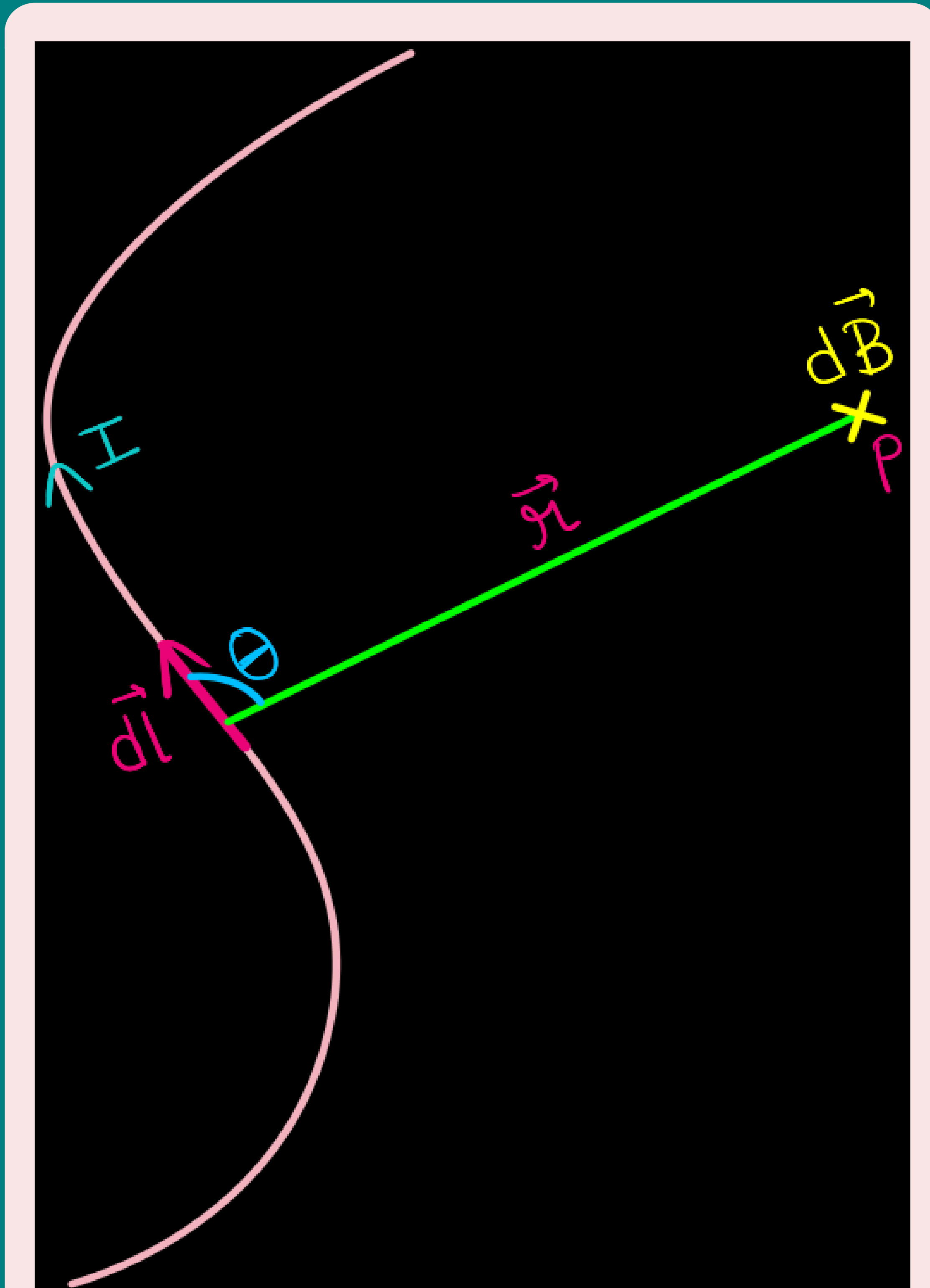


Figure: Magnetic Field around a Wire and its direction is give by right hand thumb rule. Source : Java Lab

Observations and Results:

- **Iron filings** sprinkled around the wire **arrange themselves in concentric circles** with the wire as the center.
- The **magnetic field** is the **area surrounding a magnet** in which the **magnetic force exists**.
- **Oersted concluded** that **moving charges or currents produced a magnetic field** in the surrounding space.

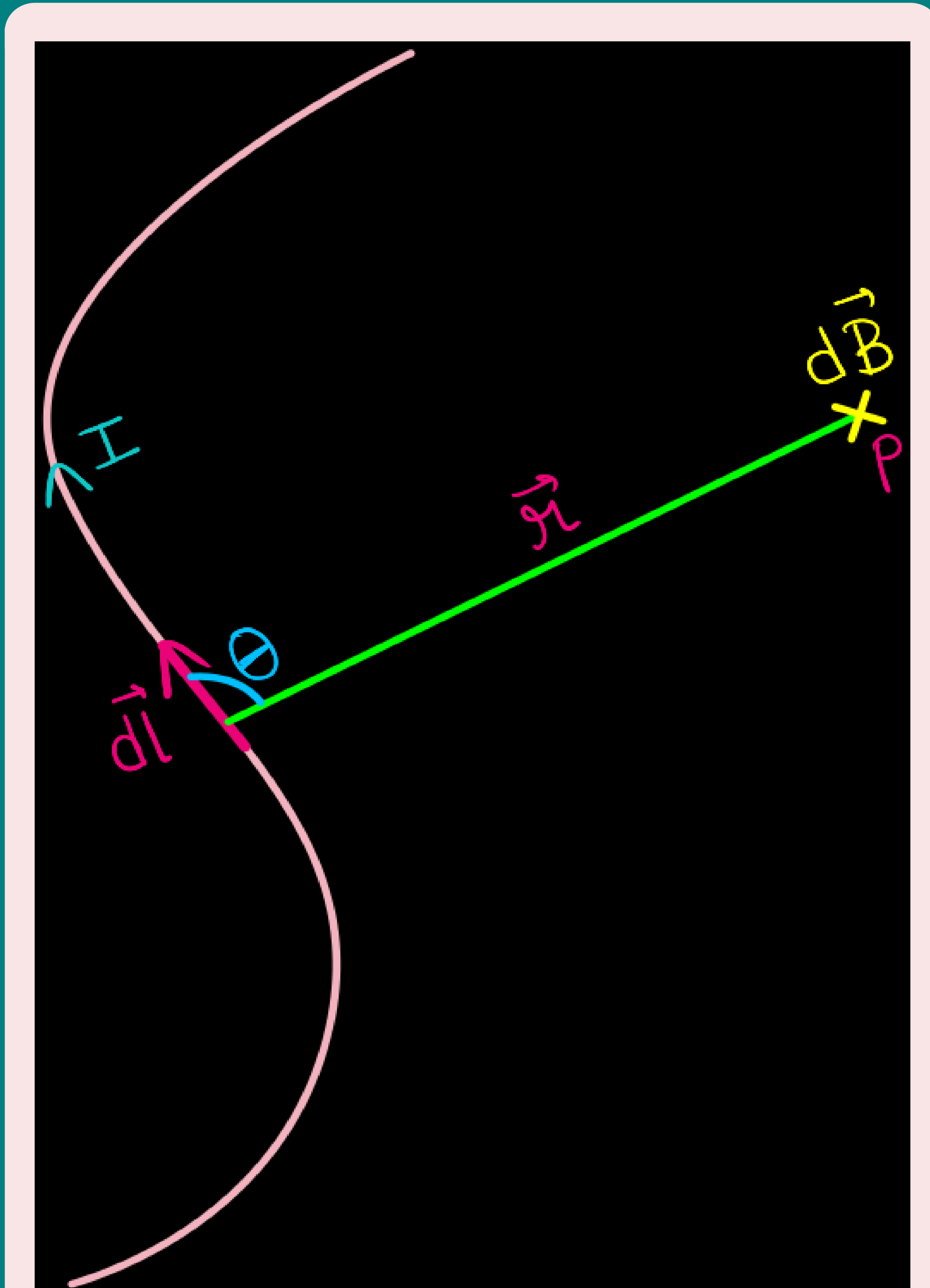
BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- Figure shows a finite conductor carrying **current I** .
- Consider an **infinitesimal element $d\vec{l}$** of the conductor.
- The **magnetic field $d\vec{B}$** due to this element is to be determined **at a point P** which is at a **distance r** from it. Let θ be the **angle between $d\vec{l}$ and the displacement vector r** .

BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- According to Biot-Savart's law, the magnitude of the magnetic field $d\vec{B}$ is proportional to

- 1 The current I ,

$$(d\vec{B} \propto I)$$

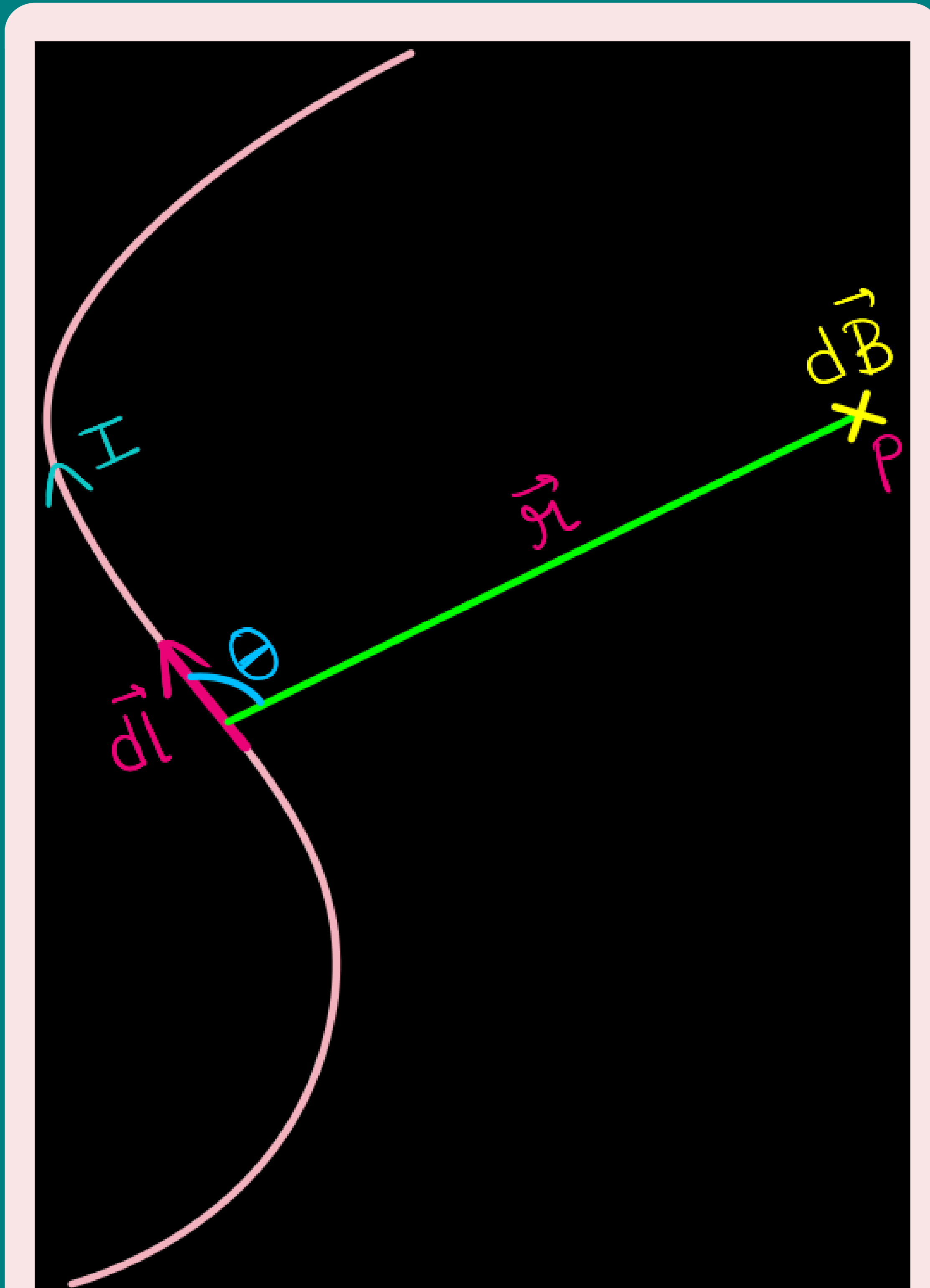
- 2 The element length dl ,

$$(d\vec{B} \propto d\vec{l}) \quad \text{and}$$

- 3 Inversely proportional to the square of the distance r

$$\left(d\vec{B} \propto \frac{1}{r^2}\right).$$

BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- **direction** of $d\vec{B}$ is **perpendicular to the plane** containing $d\vec{l}$ and $\vec{r}$. Thus, in vector notation,

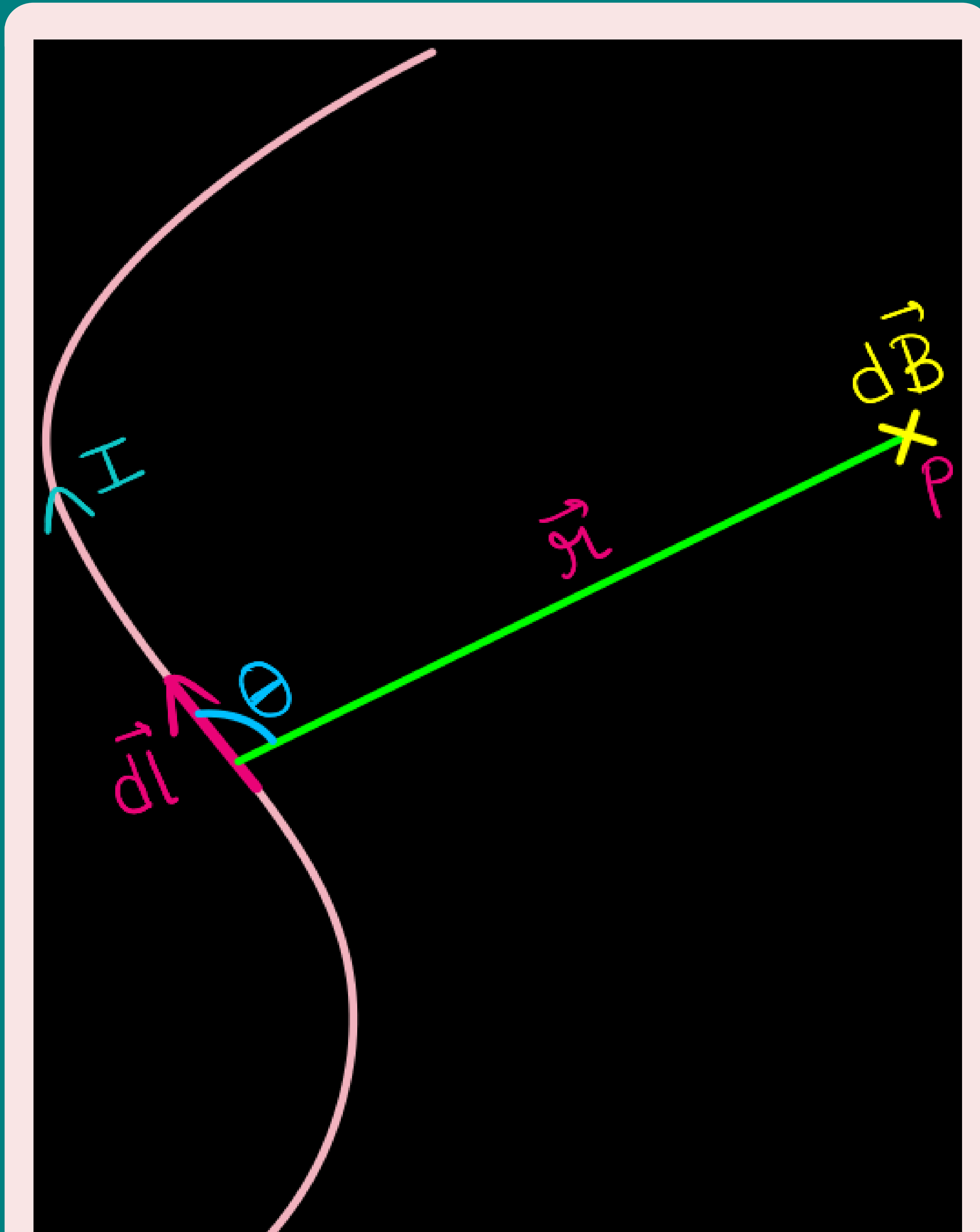
$$d\vec{B} \propto \frac{I d\vec{l} \times \hat{r}}{r^2}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

or

$$|d\vec{B}| = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- where $\frac{\mu_0}{4\pi}$ is a **constant of proportionality** in SI units has the exact value.

$$\frac{\mu_0}{4\pi} = 10^{-7} \text{ Tm/A}$$

- We call μ_0 the **permeability of free space (or vacuum)**.

Similarities and Differences b/w Biot-Savart and Coulomb's law

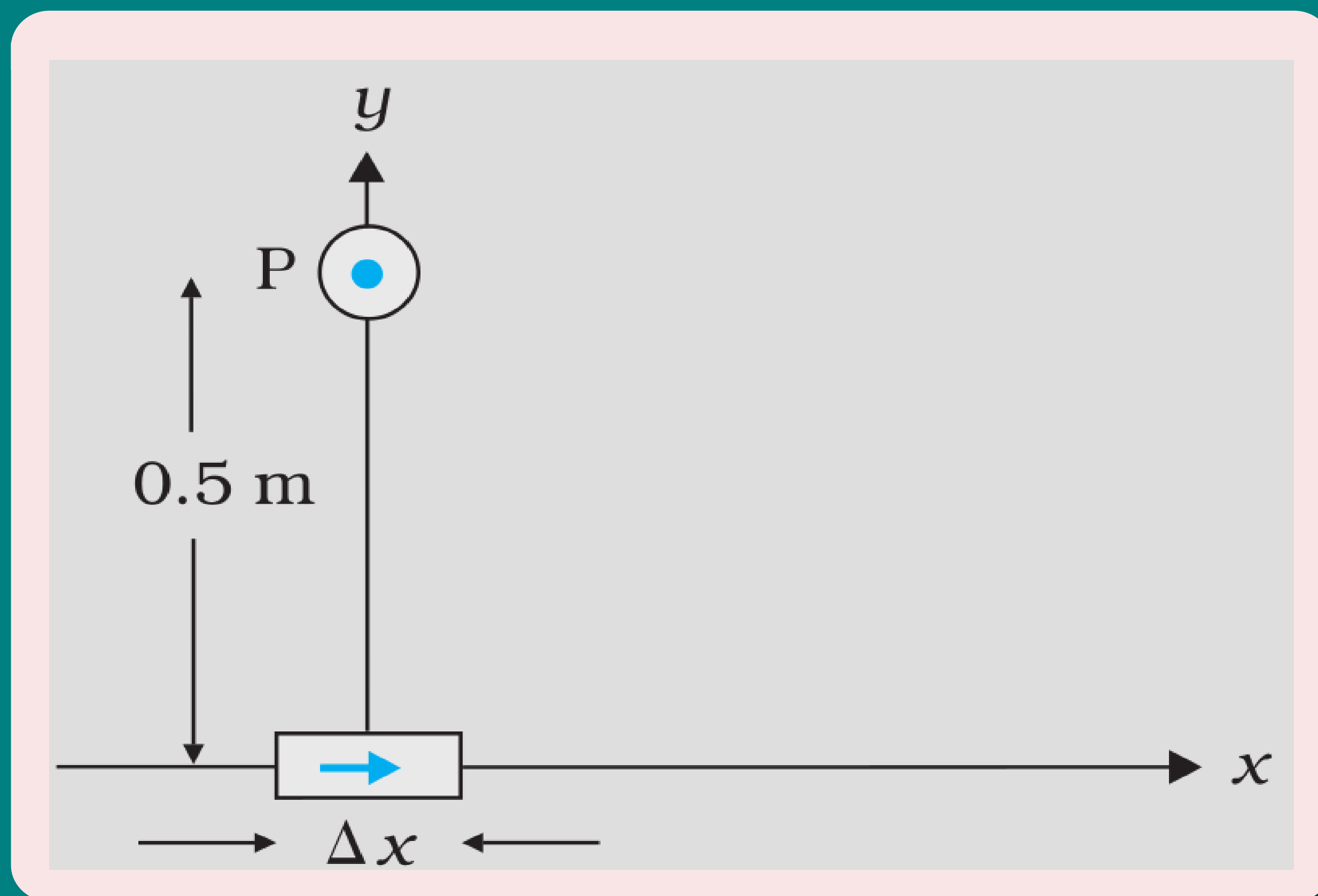
- **Both are long range**, since **both depend inversely** on the **square of distance** from the source to the point of interest.
- The **principle of superposition applies to both** fields.
- The **electrostatic field is produced by a scalar source**, namely, the electric charge. The **magnetic field is produced by a vector source** $I d\vec{l}$.
- The **electrostatic field is along the displacement vector joining the source and the field point**. The **magnetic field is perpendicular to the plane containing the displacement vector $\vec{r}$ and the current element $I d\vec{l}$** .
- There is an **angle dependence in the Biot-Savart law** which is **not present in the electrostatic case**.

Relation between Permittivity of free space (ϵ_0) and the permeability of free space (μ_0)

- There is an **interesting relation** between ϵ_0 , the **permittivity of free space**; μ_0 , the **permeability of free space**; and c , the **speed of light in vacuum**:

$$\begin{aligned}\epsilon_0 \times \mu_0 &= 4\pi\epsilon_0 \times \frac{\mu_0}{4\pi} \\ \epsilon_0 \times \mu_0 &= \frac{1}{9 \times 10^9} \times 10^{-7} \\ \epsilon_0 \times \mu_0 &= \frac{1}{9 \times 10^{16}} = \frac{1}{(3 \times 10^8)^2} \\ \epsilon_0 \times \mu_0 &= \frac{1}{c^2} \quad \boxed{c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}}\end{aligned}$$

Example 4.5



An element $\Delta l = \Delta x \hat{i}$ is placed at the origin and carries a large current $I = 10 \text{ A}$. What is the magnetic field on the y-axis at a distance of 0.5 m . $\Delta x = 1 \text{ cm}$.

Applications of Biot-Savart Law

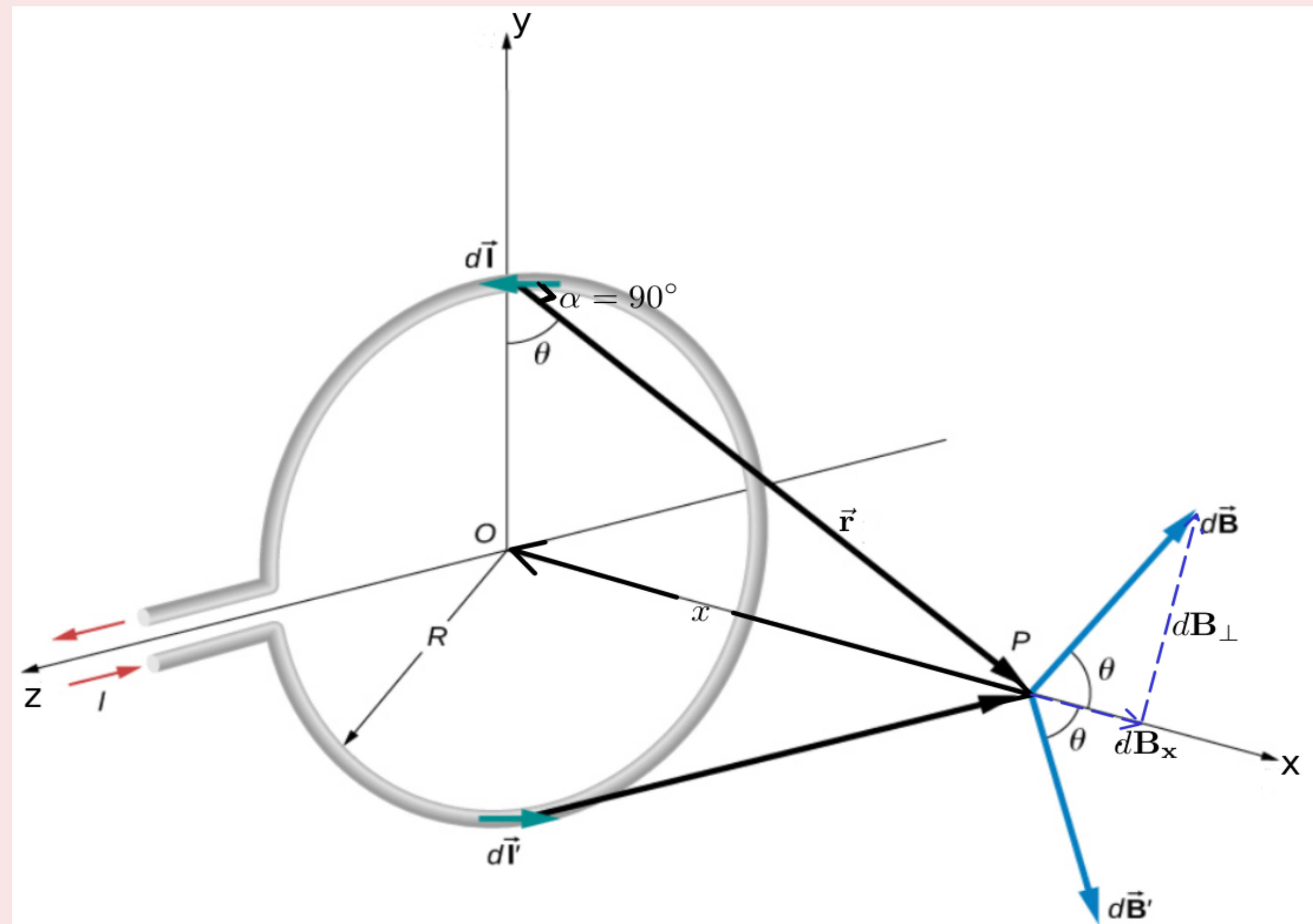


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- Figure depicts a **circular loop carrying a steady current I** .
- The **loop is placed in the y - z plane** with its center at the **origin O** and has a **radius R** .
- **We wish to calculate the magnetic field at the point P on this axis.**
- **Let x be the distance of P from the center O of the loop.**

Applications of Biot-Savart Law

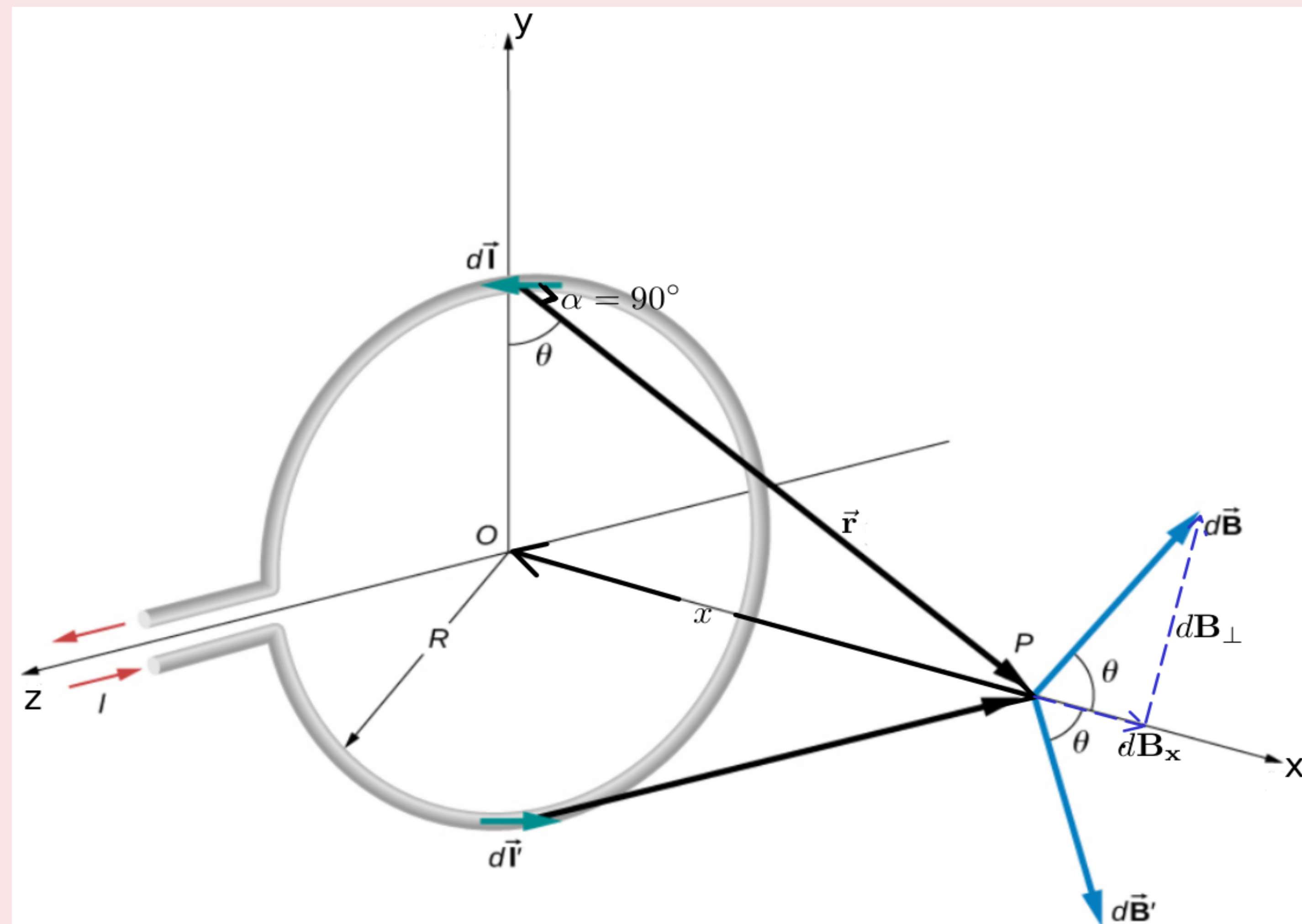


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- The magnitude dB of the magnetic field due to $d\vec{l}$ is given by the Biot-Savart law

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \alpha}{r^2}$$
$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \alpha}{(x^2 + R^2)} \quad (\because r^2 = R^2 + x^2)$$

- The element $d\vec{l}$ will be perpendicular to the displacement vector $\vec{r}$ (Or $\alpha = 90^\circ$)

Applications of Biot-Savart Law

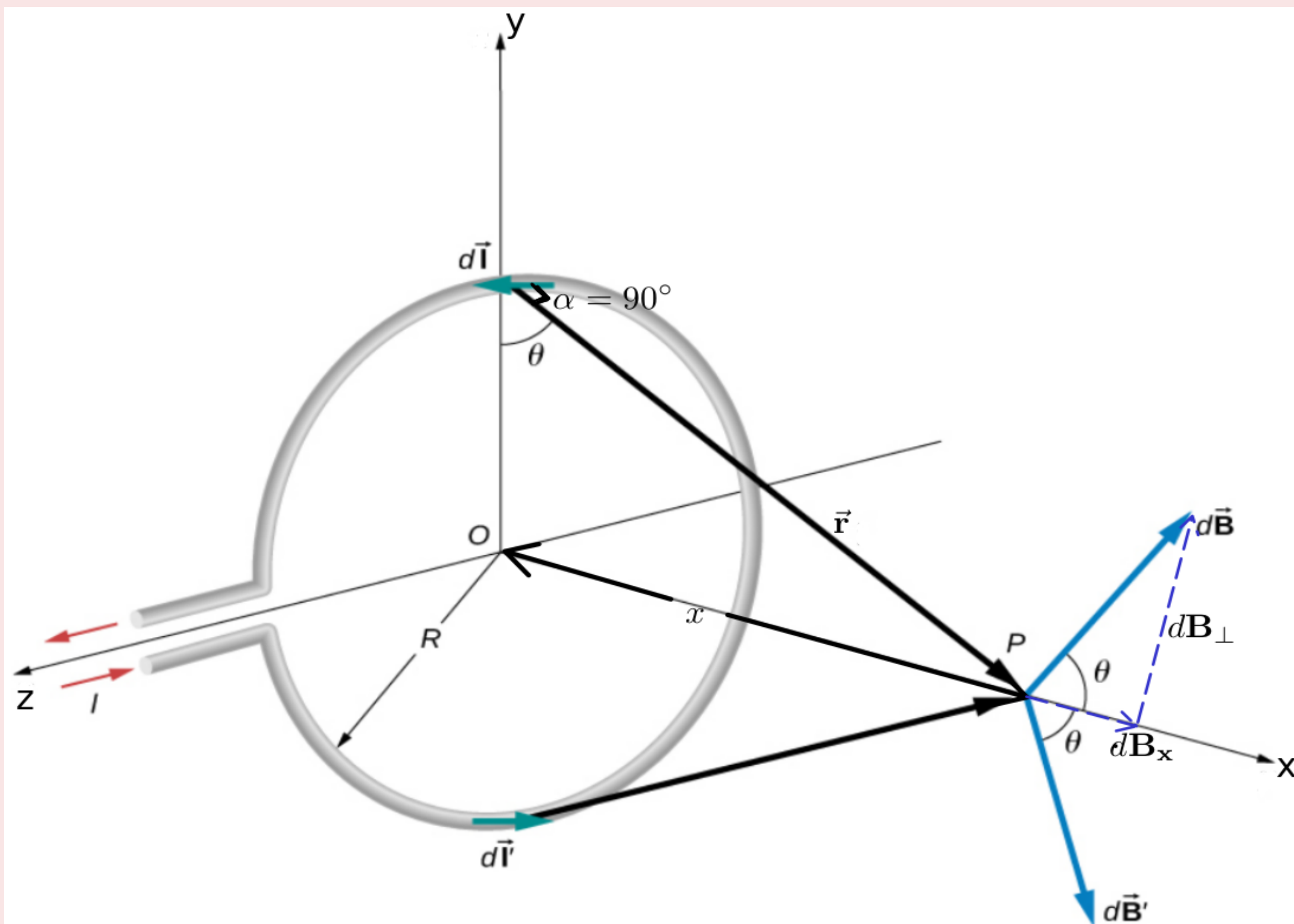


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- Thus,

$$dB = \frac{\mu_0}{4\pi} \frac{I dl}{(x^2 + R^2)^{3/2}} \quad \dots\dots(1)$$

- $d\vec{B}$ has an **x-component** dB_x and a **component perpendicular** to x-axis, $dB_{\perp}$.
- When the **perpendicular components** ($dB_{\perp}$) are **summed over, they cancel out**.
- Thus, only the x-component (dB_x) survives

Applications of Biot-Savart Law

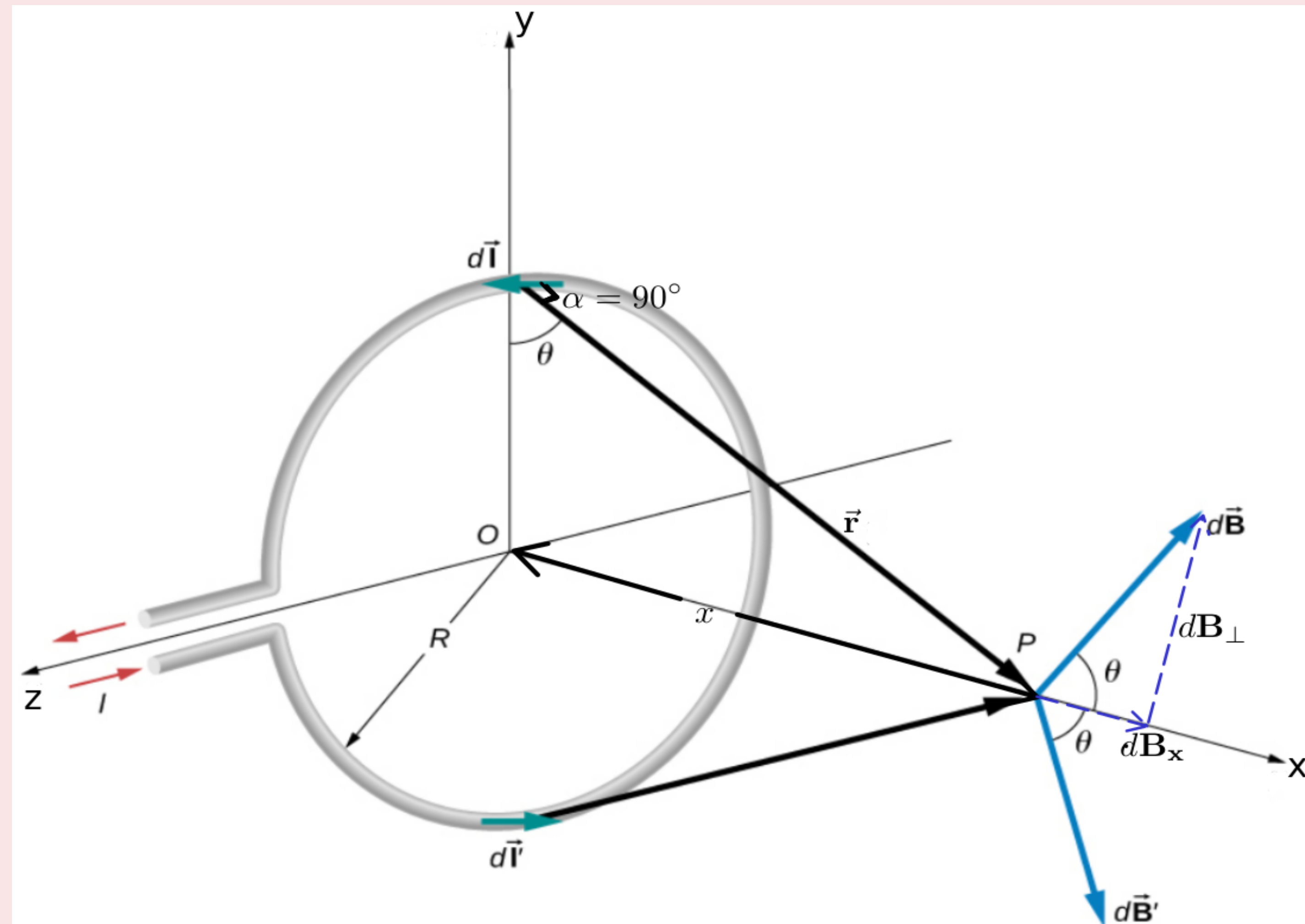


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- The **net contribution along x-direction** can be **obtained by integrating** $dB_x = dB \cos \theta$ over the loop.

$$\cos \theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}} \quad \dots\dots(2)$$

- From Eqs. (1) and (2),

$$dB_x = \frac{\mu_0}{4\pi} \frac{Idl}{(x^2 + R^2)} \times \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

Applications of Biot-Savart Law

Magnetic Field On The Axis Of a Circular Current Loop

- $$dB_x = \frac{\mu_0}{4\pi} \frac{I R dl}{(x^2 + R^2)^{\frac{3}{2}}}$$

- Thus, the **magnetic field at P due to entire circular loop** is

$$\vec{B} = B_x \hat{i} = \int_0^{2\pi R} \frac{\mu_0}{4\pi} \frac{I R dl}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{I R \times 2\pi R}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i} \text{ Or } \boxed{\vec{B} = \frac{\mu_0}{2} \frac{I R^2}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}}$$

Applications of Biot-Savart Law

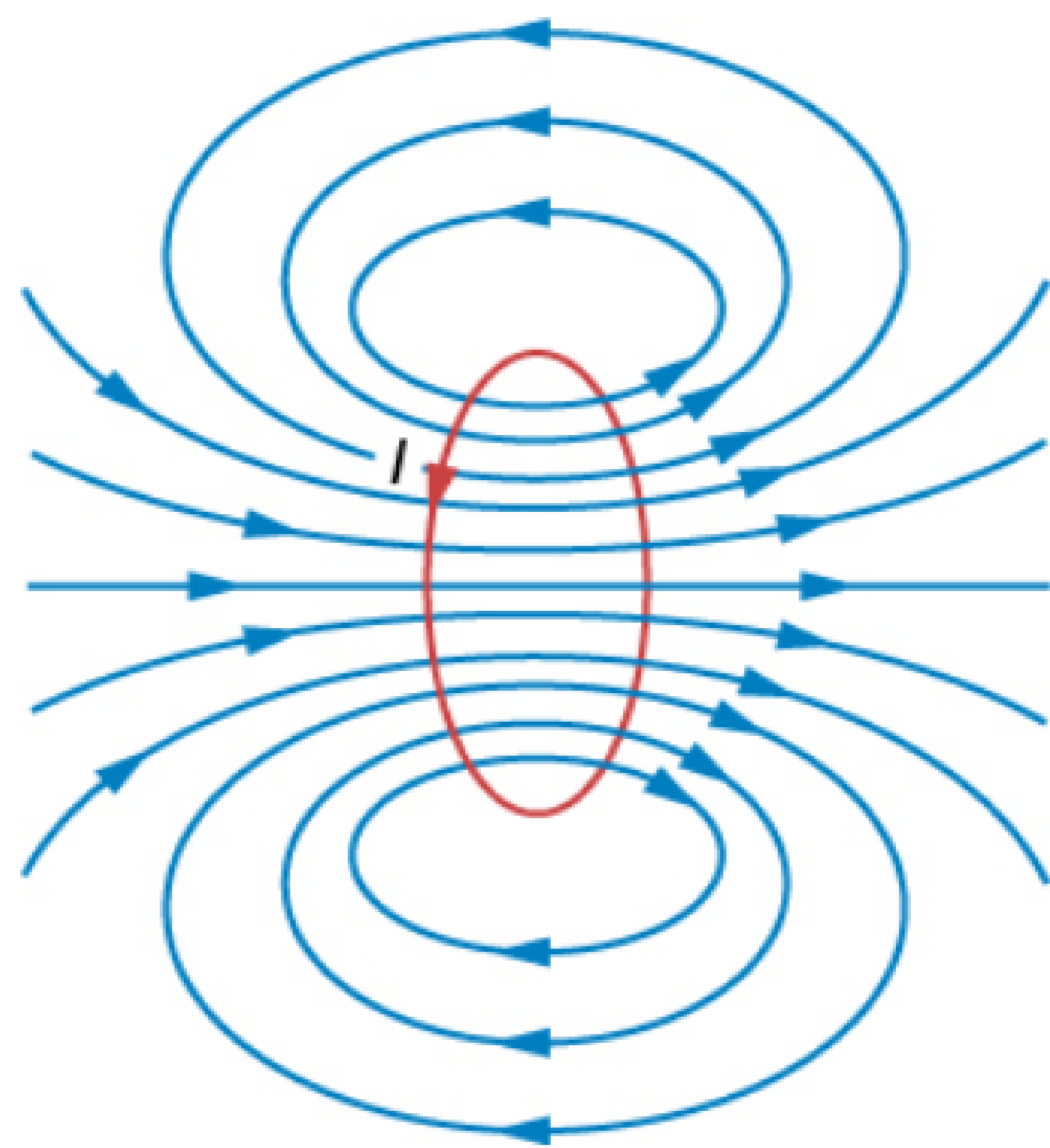


Figure 12.12 Sketch of the magnetic field lines of a circular current loop.

Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The **field at the center of the loop** ($x = 0$).

$$\vec{B} = \frac{\mu_0}{2} \frac{I R^2}{(0 + R^2)^{\frac{3}{2}}} \hat{i}$$

$$\boxed{\vec{B} = \frac{\mu_0}{2} \frac{I}{R} \hat{i}}$$

- The **magnetic field lines** due to a circular wire **form closed loops**

Example 4.6

A straight wire carrying a **current of 12 A** is bent into a semi-circular arc of **radius 2.0 cm** as shown in Fig. 4.13(a). Consider the **magnetic field $\mathbf{B}$** at the **center** of the arc.

- (a) What is the **magnetic field due to the straight segments**?
- (b) In what way the **contribution to $\mathbf{B}$** from the **semicircle differs from that of a circular loop** and in what way does it resemble?
- (c) Would your answer be different **if the wire were bent** into a semi-circular arc of the same radius but **in the opposite way** as shown in Fig. 4.13(b)?

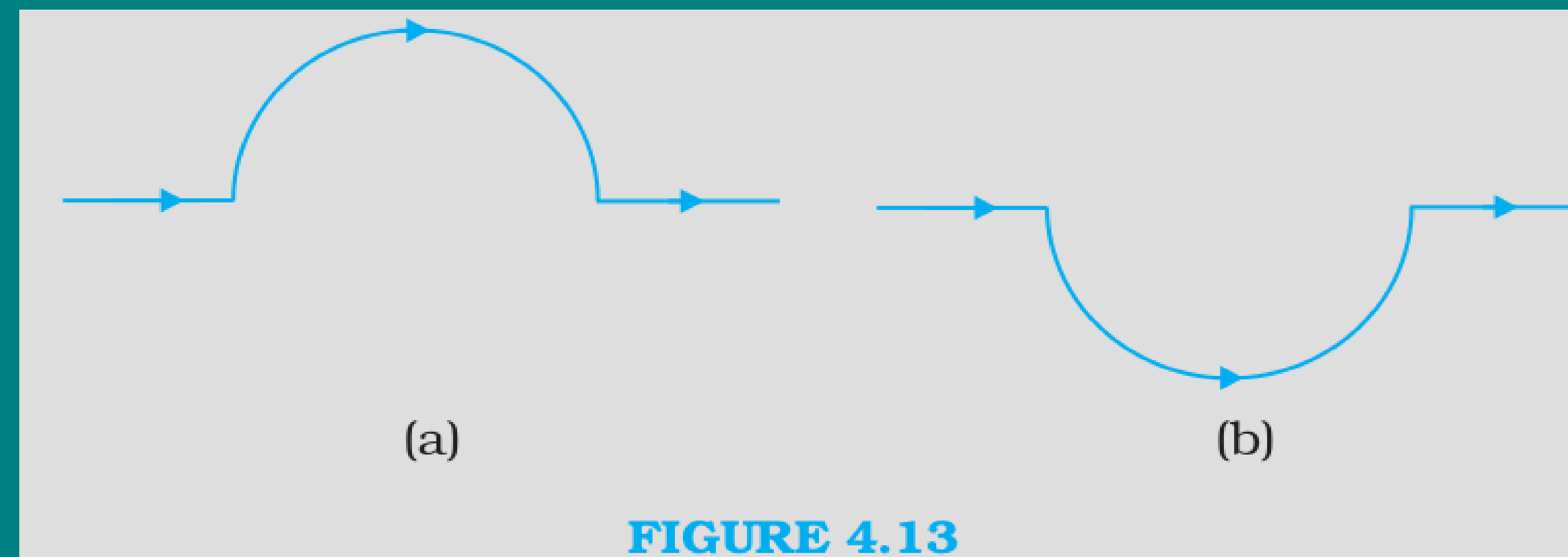


FIGURE 4.13

Example 4.7

Consider a tightly wound **100 turn** coil of **radius 10 cm**, carrying a **current of 1 A**.
What is the magnitude of the magnetic field at the center of the coil?